Short Question Assignment: Day-05

Question 1: The meaning of Zero:

you Analyze a loss function $L(w_1, w_2)$. At a specific point $w_1 = 5, w_2 = -3$. you calci the partial derivates and find that

$$\frac{\partial L}{\partial w_1} = 0 \quad , \quad \frac{\partial L}{\partial w_2} = 0.$$

Geometricly, what does this tell you in loss function explin?

Solution: The loss function is perfectly flat at the exact coordinate. This is known a critical point.

Question 2: Gradient dimensionality

you are training a LLM. The model's loss fn $L(w)$ depends on a massive weight w vector, containing 1.5 billion parameters. What the exact dimensions (shape) of the gradient vector ∇L?

Solution: The gd vector will have exactly 1.5b dimension

Shape $(1.5b, 1)$.

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Qustion 3: The Direction of Decent

a loss function $\nabla L = \begin{bmatrix} 4.2 \\ -1.8 \end{bmatrix}$

update your weight and decrese the loss. Should you move in the direction of vector $[4.2, -1.8]^T$ or $[-4.2, 1.8]^T$. explain why?

Solution:

You should move $[-4.2, 1.8]^T$. The calculated gradient points uphill. To decrese the loss.

Qustin 4: Orthogonality and counter maps:

Imagine looking down at a 2D contour map of a loss function, where each ring represents a speific error value.

If you evalate the gradient vector at a speific point on one of these counters line, what is the geometric angle Betn gradient vector and counter line itself?

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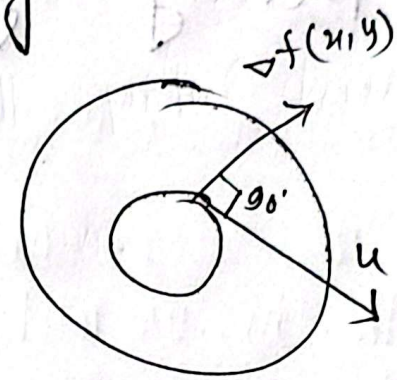
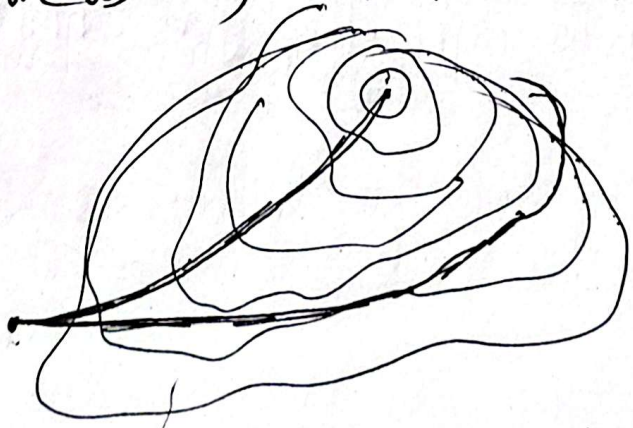
Solution: The angle is exactly 90°. The steepest path up a mountain is always perfectly perpendicular to the flat contour paths wrapping around it.



Question 5: The Directional Derivative

Suppose you are standing on the side of hill. The gd vector points straight up the steepest part of the hill.

If you decide to walk in a direction that is exactly 90° (orthogonal) to the gradient vector, what happens to your ~~evaluation~~ elevation (your (x,y) value) in that immediate instant?



$D_u f = 0.$

একই লেভেল বোঝাবে height তিনি same.

Solution: your উচ্চতা elevation (boss value) would not change.

If you walk 90° away from the steepest path, you are walking perfectly along a counter line where the slope is 0.